

# Basic of Polynomials

Let  $S$  be one of the sets  $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ .

- **Definition 1**

A **polynomial** with coefficients in  $S$  is a formal expression

$$p(x) = a_0 + a_1x + \cdots + a_nx^n$$

where  $a_0, a_1, \dots, a_n \in S$  and  $n$  is a nonnegative integer.

The set of all polynomials with coefficients in  $S$  is denoted by  $S[x]$ .

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- **Definition 2**

Let  $p(x) = a_0 + a_1x + \cdots + a_nx^n$ . If  $a_n \neq 0$ , we say that the **degree** of  $p(x)$  is  $n$  and is denoted

$$\deg(p(x)) = n.$$

Moreover, we say that the **leading coefficient** of  $p(x)$  is  $a_n$ .

- **Example 2.1**

$$p(x) = 2 + x - 3x^2 + 7x^4 \in \mathbb{Z}[x]$$

$$p(x) = 1 + \frac{2}{3}x - \frac{5}{2}x^3 \in \mathbb{Q}[x]$$

$$p(x) = x + \sqrt{2}x^2 - \frac{1}{7}x^5 \in \mathbb{R}[x]$$

$$p(x) = i - 2x + (i+1)x^2 \in \mathbb{C}[x]$$

- **Remark 2.2**

The degree of the zero polynomial  $0(x) = 0$  is undefined.

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- **Definition 3**

A polynomial  $p(x) \in S[x]$  is called **monic** if its leading coefficient is 1.

- **Example 3.1**

Let  $p(x) = 1 - 5x^2 + \frac{\sqrt{2}}{2}x^7 + x^9 \in \mathbb{R}[x]$ , then  $p(x)$  is a monic polynomial with  $\deg(p(x)) = 9$ .

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- **Definition 4**

Two polynomials  $p(x), q(x) \in S[x]$  are **equal** if  $\deg(p(x)) = \deg(p(x))$  (or  $\deg(q(x))$  if defined) and their corresponding coefficients are equal.

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- **Definition 5**

Let  $p(x) = a_0 + a_1x + \cdots + a_nx^n, q(x) = b_0 + b_1x + \cdots + b_mx^m$ . Then

$$p(x) + q(x) = \sum_{k=0}^{\max(n,m)} (a_k + b_k) x^k.$$

$$p(x)q(x) = \sum_{k=0}^{n+m} c_k x^k \quad \text{where} \quad c_k = \sum_{j=0}^k a_j b_{k-j}.$$

- **Example 5.1**

Let  $p(x) = 5 + 2x + 6x^2$  and  $q(x) = 3 + x + 4x^2 + 5x^3$ , then

$$\begin{array}{cccccc} a_0 = 5, & a_1 = 2, & a_2 = 6, & a_3 = 0, & a_4 = 0, & a_5 = 0 \\ b_0 = 3, & b_1 = 1, & b_2 = 4, & b_3 = 5, & b_4 = 0, & b_5 = 0. \end{array}$$

Then

$$\begin{aligned} p(x) + q(x) &= \sum_{k=0}^3 (a_k + b_k) x^k \\ &= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3 \\ &= (5 + 3) + (2 + 1)x + (6 + 4)x^2 + (0 + 5)x^3 \\ &= 8 + 3x + 10x^2 + 5x^3. \end{aligned}$$

$$\begin{aligned} p(x)q(x) &= \sum_{k=0}^5 c_k x^k \quad \text{where} \quad c_k = \sum_{j=0}^k a_j b_{k-j} \\ &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 \end{aligned}$$

We compute each coefficient  $c_k$  directly as follows:

- $c_0 = a_0b_0 = 5 \cdot 3 = 15.$
- $c_1 = a_0b_1 + a_1b_0 = 5 \cdot 1 + 2 \cdot 3 = 5 + 6 = 11.$
- $c_2 = a_0b_2 + a_1b_1 + a_2b_0 = 5 \cdot 4 + 2 \cdot 1 + 6 \cdot 3 = 20 + 2 + 18 = 40.$
- $c_3 = a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0 = 5 \cdot 5 + 2 \cdot 4 + 6 \cdot 1 + 0 \cdot 3 = 25 + 8 + 6 + 0 = 39.$
- $c_4 = a_0b_4 + a_1b_3 + a_2b_2 + a_3b_1 + a_4b_0 = 5 \cdot 0 + 2 \cdot 5 + 6 \cdot 4 + 0 \cdot 1 + 0 \cdot 3 = 10 + 24 = 34.$
- $c_5 = a_0b_5 + a_1b_4 + a_2b_3 + a_3b_2 + a_4b_1 + a_5b_0 = 5 \cdot 0 + 2 \cdot 0 + 6 \cdot 5 + 0 + 0 + 0 = 30.$

Therefore, the product polynomial is:

$$p(x)q(x) = 15 + 11x + 40x^2 + 39x^3 + 34x^4 + 30x^5.$$

◦ The arithmetic properties of  $S[x]$  are as follows:

1.  $(p(x) + q(x)) + r(x) = p(x) + (q(x) + r(x)).$
2.  $p(x) + 0(x) = p(x).$
3. For each  $p(x) \in S[x]$ , there exists  $-p(x) \in S[x]$  such that  $p(x) + (-p(x)) = 0(x).$
4.  $p(x) + q(x) = q(x) + p(x).$
5.  $(p(x)q(x))r(x) = p(x)(q(x)r(x)).$
6.  $p(x) \cdot 1 = p(x).$
7.  $p(x)q(x) = q(x)p(x).$
8.  $p(x)(q(x) + r(x)) = p(x)q(x) + p(x)r(x).$

■ **Remark 5.2**

Let  $p(x), q(x) \in S[x]$  be nonzero.

1. If  $p(x) + q(x) \neq 0(x)$ , then

$$\deg(p(x) + q(x)) \leq \max \{ \deg(p(x)), \deg(q(x)) \}.$$

- 2.

$$\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x)).$$

■ **Question 5.3**

Can the inequality in Remark 5.2(1) be strict even when both polynomials are nonzero?

■ **Solution 5.3.1**

Yes. Note that for  $p(x) = 1 + x - x^2$  and  $q(x) = 2x + x^2$ , we have  $p(x) + q(x) = 1 + 3x$ .

Here,  $\deg(p(x) + q(x)) = 1 < \max\{2, 2\} = 2$ .

We use  $\mathbb{K}$  to denote  $\mathbb{Q}, \mathbb{R}$ , or  $\mathbb{C}$ .

■ **Remark 5.4**

Let  $p(x), q(x) \in \mathbb{K}[x]$ . If  $p(x)q(x) = 0$ , then  $p(x) = 0 \vee q(x) = 0$ . In fact, suppose for the sake of contradiction that  $p(x)q(x) = 0$  and  $p(x) \neq 0 \wedge q(x) \neq 0$ . Hence we obtain that:

$$\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x)).$$

But since  $p(x)q(x) = 0$ , the degree of the left side is undefined, which yields a contradiction.

■ **Theorem 5.5 (The division algorithm for  $\mathbb{K}[x]$ )**

Let  $p(x), s(x) \in \mathbb{K}[x]$  with  $s(x) \neq 0$ , then there exist unique polynomials  $q(x), r(x) \in \mathbb{K}[x]$  such that

$$p(x) = s(x)q(x) + r(x)$$

with either  $r(x) = 0$  or  $\deg(r(x)) < \deg(s(x))$ .

■ **Proof 5.5.1**

**Existence**

If  $p(x) = 0$ , then we take  $q(x) = r(x) = 0$ .

If  $p(x) \neq 0$ , we use induction on  $\deg(p(x))$ .

- **Base step:** Suppose that  $\deg(p(x)) = 0$ . Since  $s(x) \neq 0$ , we have the following cases:

- If  $\deg(s(x)) = 0$ , then  $s(x) = b \neq 0$ , so  $b^{-1} \in \mathbb{K}$ . We take  $q(x) = b^{-1}p(x)$  and  $r(x) = 0$ .

- If  $\deg(s(x)) > 0$ , we take  $q(x) = 0$  and  $r(x) = p(x)$ , where  $\deg(r(x)) = 0 < \deg(s(x))$ .

Thus, the base step holds.

- **Inductive step:** Suppose that the result is true for any polynomial with degree less than or equal to  $n - 1$ . Now we need to check if the result is true for  $\deg(p(x)) = n$ . Let  $\deg(p(x)) = n$  and  $\deg(s(x)) = m$ . We consider two cases:

- If  $n < m$ , we take  $q(x) = 0$  and  $r(x) = p(x)$ .
- If  $n \geq m$ , let  $p(x) = a_0 + a_1x + \dots + a_nx^n$  and  $s(x) = b_0 + b_1x + \dots + b_mx^m$  with  $a_n, b_m \neq 0$ . We construct the difference:

$$\begin{aligned} p(x) - b_m^{-1}a_nx^{n-m}s(x) &= (a_0 + a_1x + \dots + a_nx^n) - b_m^{-1}a_nx^{n-m}(b_0 + b_1x + \dots \\ &= (a_0 + a_1x + \dots + a_nx^n) - (b_m^{-1}a_nb_0x^{n-m} + \dots + a_nx^n) \end{aligned}$$

Notice that the leading term  $a_nx^n$  cancels out completely. That is,  $\deg(p(x) - b_m^{-1}a_nx^{n-m}s(x)) \leq n - 1$ . By the inductive hypothesis, there exist polynomials  $q'(x), r(x) \in \mathbb{K}[x]$  such that:

$$p(x) - b_m^{-1}a_nx^{n-m}s(x) = s(x)q'(x) + r(x)$$

with  $r(x) = 0$  or  $\deg(r(x)) < \deg(s(x))$ . Now, if we take  $q(x) = q'(x) + b_m^{-1}a_nx^{n-m}$ , we have:

$$p(x) = s(x)q(x) + r(x)$$

with  $r(x) = 0$  or  $\deg(r(x)) < \deg(s(x))$ . This completes the inductive step.

#### Uniqueness

Suppose that  $p(x) = s(x)q(x) + r(x)$  with  $r(x) = 0$  or  $\deg(r(x)) < \deg(s(x))$ , and  $p(x) = s(x)q'(x) + r'(x)$  with  $r'(x) = 0$  or  $\deg(r'(x)) < \deg(s(x))$ . Hence:

$$\begin{aligned} s(x)q(x) + r(x) &= s(x)q'(x) + r'(x) \\ s(x)(q(x) - q'(x)) &= r'(x) - r(x). \end{aligned}$$

If  $q(x) \neq q'(x)$ , then since  $s(x) \neq 0$ , the degree of the left side is:

$$\deg(s(x)(q(x) - q'(x))) = \deg(s(x)) + \deg(q(x) - q'(x)) \geq \deg(s(x)).$$

However, for the right side, since  $\deg(r'(x)) < \deg(s(x))$  and  $\deg(r(x)) < \deg(s(x))$ , we have:

$$\deg(r'(x) - r(x)) \leq \max\{\deg(r'(x)), \deg(r(x))\} < \deg(s(x)),$$

which yields a contradiction. Therefore,  $q(x) = q'(x)$  and thus  $r(x) = r'(x)$ .

#### ■ Example 5.6

If  $p(x) = 8x^4 - 4x^3 + 2x^2 + x + 1$  and  $s(x) = 2x^2 + 3x + 7$ , then:

$$\begin{aligned} p(x) - 4x^2s(x) &= -16x^3 - 26x^2 + x + 1 \\ -16x^3 - 26x^2 + x + 1 + 8xs(x) &= -2x^2 + 57x + 1 \\ -2x^2 + 57x + 1 + s(x) &= 60x + 8. \end{aligned}$$

Hence,  $p(x) = s(x)(4x^2 - 8x - 1) + (60x + 8)$ .

#### ○ Definition 6

Let  $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{K}[x]$ . If  $\alpha \in \mathbb{K}$ , the **evaluation** of  $p(x)$  at  $\alpha$  is:

$$p(\alpha) = a_0 + a_1\alpha + \dots + a_n\alpha^n,$$

that is,  $p(\alpha) \in \mathbb{K}$ .

#### ■ Theorem 6.1 (Remainder theorem)

Let  $p(x) \in \mathbb{K}[x]$  and let  $\alpha \in \mathbb{K}$ , then there exists  $q(x) \in \mathbb{K}[x]$  such that

$$p(x) = (x - \alpha)q(x) + p(\alpha).$$

#### ■ Proof 6.1.1

By the division algorithm, there exist unique polynomials  $q(x), r(x)$  such that:

$$p(x) = (x - \alpha)q(x) + r(x)$$

with  $r(x) = 0$  or  $\deg(r(x)) < \deg(x - \alpha) = 1$ . This implies that  $r(x)$  must be a constant polynomial, so we can write  $r(x) = c$  for some  $c \in \mathbb{K}$ .

We now evaluate the polynomial equation  $p(x) = (x - \alpha)q(x) + c$  at the point  $x = \alpha$ :

$$p(\alpha) = (\alpha - \alpha)q(\alpha) + c = 0 \cdot q(\alpha) + c = c.$$

Since  $c = p(\alpha)$ , substituting this back into the division equation yields:

$$p(x) = (x - \alpha)q(x) + p(\alpha).$$

Therefore, the theorem is proved.